

Advanced Microeconomics

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1 Recap Marginal Value of Public Funds and Value of Statistical Life

1) Answer the following questions:

- a) Briefly describe the concepts of willingness-to-pay (WTP) and Marginal Value of Public Funds (MVPF).
- b) A study reports MVPFs of changes of the Earned Income Tax Credit with values at 1.2 and 1.12, while other cash transfers have MVPFs below 1. Interpret these differences and give an intuition for why they occur.
- c) Briefly describe potential problems associated with the MVPF.

2 Application of General Equilibrium

3) Individuals A and B of a two good exchange economy (x_1, x_2) have the following utility functions:

$$U_a(x_{a1}, x_{a2}) = \ln x_{a1} + 2 \ln x_{a2}$$
$$U_b(x_{b1}, x_{b2}) = 2 \ln x_{b1} + \ln x_{b2}$$

Let the initial endowments of the individuals be $R_A = (9, 3)$ and $R_B = (12, 6)$.

- a) Find the excess demand for each good, give a definition of Walras' law and verify that the sum of the excess demands adds up to 0.
- b) Calculate the equilibrium prices of the goods.
- c) Determine the allocation that results in equilibrium.